

Section : Physics

Q.1 A charge of magnitude $3 \times 10^{-7} \text{ C}$ is located at a distance of 0.09 m from a point P. Obtain the work done in bringing a charge of $2 \times 10^{-9} \text{ C}$ from infinity to the point P.

1. $1.6 \times 10^4 \text{ J}$
2. $2.6 \times 10^{-2} \text{ J}$
3. $3.6 \times 10^{-5} \text{ J}$
4. $4.6 \times 10^5 \text{ J}$

Options 1. 1

2. 2
3. 3
4. 4

Q.2 A 12 V battery connected to a 6Ω , 10 mH coil through a switch drives a constant current in the circuit. The switch is suddenly opened. Assuming that it took 1 ms to open the switch, the average emf induced across the coil would be

1. 10 V
2. 20 V
3. 200 V
4. 12 V

Options 1. 1

2. 2
3. 3
4. 4

Q.3 Two point charges, $q_1 = 4\mu\text{C}$ and $q_2 = -3\mu\text{C}$ (with no external field) are placed at $(-6 \text{ cm}, 0, 0)$ and $(6 \text{ cm}, 0, 0)$, respectively. The amount of work required to separate the two charges infinitely away from each other will be

1. 0.9 J
2. 0.18 J
3. -0.9 J
4. -0.18 J

Options 1. 1

2. 2
3. 3
4. 4

Q.4 In an experiment with a convex lens, the length of an image is 1 cm, and the object length is 5 cm. If the object is placed at a distance of 40 cm from the lens, then the focal length of the lens is

1. 6.67 cm
2. 13.5 cm
3. 5.6 cm
4. 3.6 cm

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Match List-I with List-II

List-I

(Physical Quantity)

(A) Magnetic field

(B) Magnetic moment

(C) Pole strength

(D) Permeability of free space

List-II

(Units)

(I) J T^{-1}

(II) T m A^{-1}

(III) $\text{J T}^{-1}\text{m}^{-1}$

(IV) Wb m^{-2}

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
2. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.6 The critical angle of incidence i_c for a ray incident from a denser to a rarer medium, is that angle for which

1. the angle of reflection is 90°
2. the angle of refraction is 90°
3. the angle of refraction is 0°
4. the angle of reflection is 0°

Options 1. 1

2. 2
3. 3
4. 4

Q.7 A parallel plate capacitor having plate area 200 cm^2 and separation 2.0 mm holds a charge of $0.06 \mu\text{C}$ on applying a potential difference of 60 V . The dielectric constant of the material filled in between the plates is.

1. 0.113
2. 1.13
3. 11.3
4. 113

(Given: permittivity of the vacuum $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$)

Options 1. 1

2. 2
3. 3
4. 4

Q.8 If the net flux through a cube is $\Phi = 1.05 \text{ N m}^2 \text{ C}^{-1}$, what will be the total charge inside the cube?

(Given: The permittivity of free space is $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$).

1. $9.29 \times 10^{-11} \text{ C}$
2. $9.27 \times 10^{-10} \text{ C}$
3. $9.22 \times 10^{-6} \text{ C}$
4. $9.29 \times 10^{-12} \text{ C}$

Options 1. 1

2. 2
3. 3
4. 4

Q.9 A battery of emf 12 V and internal resistance 3Ω is connected to an external resistor R .

If the current in the circuit is 0.6 A , the voltage across the external resistor will be

1. 10.2 V
2. 17.0 V
3. 12.0 V
4. 13.8 V

Options 1. 1

2. 2
3. 3
4. 4

Q.10 A long wire with a small current element of length 1 cm is placed at the origin and carries a current of 10 A along the x -axis. The magnitude of the magnetic field, due to the element, on the y -axis at a distance $r = 0.5 \text{ m}$ from it, would be

1. $4 \times 10^{-8} \text{ T}$

$2.5 \times 10^{-8} \text{ T}$

$3.6 \times 10^{-8} \text{ T}$

$4.2 \times 10^{-8} \text{ T}$

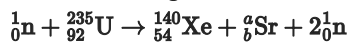
Options 1. 1

2. 2

3. 3

4. 4

Q.11 In the following nuclear reaction,



we have

1. $a = 38, b = 94$

2. $a = 94, b = 38$

3. $a = 94, b = 40$

4. $a = 96, b = 38$

Options 1. 1

2. 2

3. 3

4. 4

Q.12 The de-Broglie wavelength associated with a ball of mass $m = 150 \text{ g}$ traveling at $v = 30.0 \text{ m/s}$ would be

1. $1.47 \times 10^{-34} \text{ m}$

2. $14.7 \times 10^{-34} \text{ m}$

3. $0.147 \times 10^{-34} \text{ m}$

4. $7.14 \times 10^{-34} \text{ m}$

Options 1. 1

2. 2

3. 3

4. 4

Q.13 A bar magnet of magnetic moment 5.0 A m^2 has poles 20 cm apart. The pole strength would be

1. 20 A m

2. 30 A m

3. 1 A m

4. 25 A m

Options 1. 1

2. 2

3. 3

4. 4

Q.14 Which of the following statements are correct?

(A) The electrostatic repulsive force between the protons can be greater than the nuclear force to bind the nucleons together inside a nucleus.

(B) The repulsive electrostatic force between protons in smaller nuclei is much smaller than the nuclear force between nucleons inside a nucleus.

(C) The gravitational force between nucleons is much smaller than the nuclear force between the nucleons inside a nucleus.

(D) The binding energy per nucleon between nucleons is almost constant because the nuclear force is a long range force.

Choose the correct answer from the options given below:

1. (A) and (D) only

2. (B) and (C) only

3. (C) and (D) only

4. (A) only

Options 1. 1

2. 2

3. 3

Q.15 A resistor develops 800 J of thermal energy in 20 s on applying a potential difference of 20 V. Its resistance is

1. 20 Ω

2. 10 Ω

3. 40 Ω

4. 0.5 Ω

Options 1. 1

2. 2

3. 3

4. 4

Q.16 An inductor of 500 mH is in series with a resistance and a variable capacitor connected to a source of frequency 0.4 kHz. The value of capacitance of the capacitor to get a maximum current will be

1. 2.3 μF

2. 0.32 μF

3. 63 μF

4. 0.62 μF

Options 1. 1

2. 2

3. 3

4. 4

Q.17

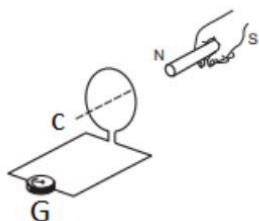


Figure shows a coil C connected to a galvanometer G. When the North-pole of a bar magnet is pushed towards the coil, the pointer in the galvanometer deflects. Regarding this set up, the following statements are given:

(A) It indicates the presence of electric current in the coil.

(B) The deflection is found to be smaller when the magnet is pushed towards the coil faster.

(C) There is repulsion in the moving magnet and the magnetic pole induced in the coil facing towards the N pole of the magnet.

(D) If the bar magnet does not move, there is no induced current in the coil.

Choose the **correct** answer from the options given below:

1. (A), (C) and (D) only

2. (A), (B) and (C) only

3. (A), (B), (C) and (D)

4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.18 If focal length of a concave lens is 50 cm, then the power of the lens would be

1. +5 D

2. - 5 D

3. +2 D

4. - 2 D

Options 1. 1

2. 2
3. 3
4. 4

Q.19 Which of the following statement(s) is/are true for p-type semiconductors?

- (A) Holes are minority carriers and pentavalent atoms are the dopants.
- (B) Electrons are majority carriers and trivalent atoms are the dopants.
- (C) Holes are majority carriers and trivalent atoms are the dopants.
- (D) Electrons are minority carriers and pentavalent atoms are the dopants.

Choose the correct answer from the options given below:

1. (B) only
2. (C) only
3. (C) and (D) only
4. (A) and (C) only

Options

1. 1
2. 2
3. 3
4. 4

Q.20 Which of the following statements are correct?

- (A) All mirrors follow the laws of reflection.
- (B) The angle between the ray of incidence and the plane surface of the mirror is equal to the angle between the plane surface of mirror and the ray of reflection for plane mirror
- (C) the rays coming parallel to the principal axis will go after reflection through the focus of the curved mirror
- (D) the rays coming to the pole of a curved mirror making an angle with axis will be reflected making the equal angle with the axis on the other side of the axis.

Choose the correct answer from the options given below:

1. (A), (B), (C) and (D)
2. (A), (C) and (D) only
3. (A), (B) and (C) only
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.21 In the (i) absence of electric field, and in the (ii) presence of electric field, the paths of electrons between successive collisions with the positive ions of the metal, are

1. (i) Straight line, (ii) straight line
2. (i) Curved, (ii) straight line
3. (i) Curved, (ii) curved
4. (i) Straight line, (ii) Curved in general

Options

1. 1
2. 2
3. 3
4. 4

Q.22 Match List-I with List-II

List-I

List-II

(Electromagnetic waves) (Wavelength range)

- | | |
|--------------------|----------------------------|
| (A) X-rays | (I) 1 mm to 700 nm |
| (B) Radio waves | (II) 0.1 m to 1 mm |
| (C) Infrared waves | (III) 1 nm to 10^{-3} nm |
| (D) Microwaves | (IV) >0.1 m |

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4

Q.23 Resistivity of a conductor depends on

1. its material and the dimensions of the conductor
2. its material and temperature of the conductor
3. the dimensions of the conductor only
4. the temperature of the conductor only

Options 1. 1

2. 2
3. 3
4. 4

Q.24 Conductors develop electric currents in them

1. on applying electric field
2. on placing the conductor in the magnetic field
3. on applying gravitational field only
4. on applying the magnetic field and gravitational field

Options 1. 1

2. 2
3. 3
4. 4

Q.25 A conductor is placed along z-axis carrying current in z direction in uniform magnetic field directed along y-axis. The magnetic force acting on the conductor is directed along:

1. positive x-axis
2. positive y-axis
3. positive z-axis
4. negative x-axis

Options 1. 1

2. 2
3. 3
4. 4

Q.26 The average emf induced in a coil is 2 V when current is changed in 0.4 s

- (A) from 5 A to 2 A and the self-inductance of the coil is 0.266 mH
(B) from 4 A to 4 A in the opposite direction, the self-inductance of the coil is 0.10 mH

1. (A) is correct, (B) is incorrect
2. Both (A) and (B) are correct
3. (A) is incorrect, (B) is correct
4. Both (A) and (B) are incorrect

Options 1. 1

2. 2
3. 3
4. 4

Q.27 Which of the following statement(s) is/are correct?

- (A) The power of a lens is the ability of the lens to converge or diverge the incident rays.
(B) S.I unit of the power of a lens is dioptre while focal length is in centimetres
(C) For a lens of larger focal length, power is smaller
(D) In any combination of lenses, the power of combination is not algebraic addition of power of combined lenses

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (B), (C) and (D) only
3. (A) and (B) only
4. (A) only

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Which of the following statements are correct?

- (A) Total internal reflection occurs when a ray of light travels from a rarer transparent medium to a denser medium.
(B) In total internal reflection, the incident ray of light remains in the same medium after reflection.
(C) In total internal reflection, the angle of incidence inside the denser transparent medium is equal to the angle of reflection in the same medium.
(D) In total internal reflection inside a denser medium there is no angle of refraction

Choose the correct answer from the options given below:

1. (B) and (C) only
2. (A) and (B) only

3. (B), (C) and (D) only

4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.29

A proton accelerated through a potential difference of V volts has a de-Broglie wavelength λ associated with it. In order to get the same wavelength associated with an α -particle, the required accelerating potential is

1. $V/16$

2. $V/8$

3. $4V$

4. $8V$

Options 1. 1

2. 2

3. 3

4. 4

Q.30

In a Young's double-slit experiment, two slits are 1.5 mm apart while the screen is 1.2 m away. When a light of wavelength 600 nm is incident on slits, the fringe width will be

1. 0.48 mm

2. 4.5 mm

3. 4.8 mm

4. 4.2 mm

Options 1. 1

2. 2

3. 3

4. 4

Q.31

A galvanometer of resistance 520Ω is shunted with 20Ω resistance to convert it into an ammeter. The resistance of the ammeter will be

1. 16.8Ω

2. 540Ω

3. 19.3Ω

4. 18Ω

Options 1. 1

2. 2

3. 3

4. 4

Q.32

A uniform wire of resistance 12Ω is cut into three pieces in the ratio of length 1 : 2 : 3.

Now the three pieces are connected to form a triangle. A cell of emf 8 V and internal resistance 5Ω is connected across the highest of the three resistors. The current through the circuit is:

1. 4 A

2. 2 A

3. 0.5 A

4. 1 A

Options 1. 1

2. 2

3. 3

4. 4

Q.33

Which of the following statements is not correct for electromagnetic induction?

1. The magnitude of induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.

2. The magnitude of induced emf in a circuit is equal to the total change of magnetic flux through the circuit.

3. The induced emf can be increased by increasing the number of turns N of a closed coil.

4. The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it

Options 1. 1

2. 2

3. 3

4. 4

Q.34

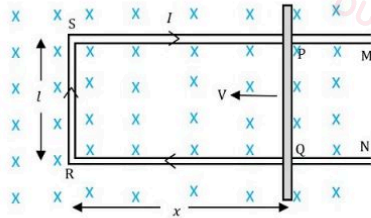


Figure shows a rectangular conductor PQRS in which the conductor PQ is free to move. The conductor PQ is moved towards the left with a constant velocity V as shown in the figure. Assume that there is no loss of energy due to friction. What will be the magnetic flux linked with the loop PQRS and the motional emf ?

1. Magnetic flux $= Bl(xV)$; Motional $emf = Blx$
2. Magnetic flux $= Blx$; Motional $emf = (BlV)/t$
3. Magnetic flux $= Blx$; Motional $emf = BlV$
4. Magnetic flux $= B/(lx)$; Motional $emf = BlV$

Options 1. 1

2. 2

3. 3

4. 4

Q.35

A polaroid sheet is rotated between two crossed polarizers. The intensity of transmitted light would be maximum, when the angle between the axes of the first polarizer and the polaroid sheet is

1. $\pi/2$

2. $\pi/4$

3. π

4. $\pi/3$

Options 1. 1

2. 2

3. 3

4. 4

Q.36

On connecting a device X in a series circuit with a battery and a resistor, a current passes through the circuit. On reversing the polarity of the battery, the current in the circuit drops to almost zero. The device X may be a

1. p - type semiconductor
2. n - type semiconductor
3. p - n junction diode
4. capacitor

Options 1. 1

2. 2

3. 3

4. 4

Q.37

Maxwell's displacement current is

1. due to flow of charges
2. due to changing gravitational field
3. due to changing electric field
4. ϵ_0 times the rate of change of magnetic flux

Options 1. 1

2. 2

3. 3

Q.38 If the protons and electrons are the only basic charges in the universe, all the observable charges have to be integral multiples of e . Thus, if an object contains x electrons and y protons, the net charge on the object will be

1. $-(x + y)e$
2. $(x + y)e$
3. $(x - y)e$
4. $(y - x)e$

Options 1. 1

2. 2
3. 3
4. 4

Q.39 For an ac source rated at 220 V, 50 Hz, which of the following statements is correct?

1. The peak value over a period of $(1/50)$ s is 220 V.
2. The average value over a period of $(1/50)$ s is 220 V.
3. The average value over a period of $(1/50)$ s is 0 V.
4. The average value over a period of $(1/50)$ s is $220\sqrt{2}$ V.

Options 1. 1

2. 2
3. 3
4. 4

Q.40 The electric field E associated with a progressive electromagnetic wave is given by $E = E_0 \sin(kx - \omega t)$. If B_0 is the amplitude of the magnetic field associated with the wave, then

1. $\frac{E_0}{B_0} = \frac{\omega}{k}$
2. $\frac{E_0}{B_0} = \frac{\omega^2}{k^2}$
3. $\frac{E_0}{B_0} = \frac{k}{\omega}$
4. $\frac{E_0}{B_0} = \frac{k^2}{\omega^2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Below given are some statements about electronic devices:

- (A) Diodes can be used for rectifying an ac voltage.
- (B) For semiconductors, band gap energy $E_g > 3$ eV.
- (C) By changing the external applied voltage, junction barriers can be changed.
- (D) p-n junction is the 'key' to all semiconductor devices.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (A), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

- Q.42 In a series combination of capacitors connected across a battery
1. each capacitor has an equal charge for certain values of capacitances only
 2. each capacitor has different charge for a certain value of its capacitance
 3. each capacitor has equal charge for any value of its capacitance
 4. each capacitor has different charge for any value of its capacitance

Options 1. 1
2. 2
3. 3
4. 4

- Q.43 A charged particle accelerated through a potential difference of V volts acquires a speed u . The particle is then made to enter perpendicularly in a uniform magnetic field B . The radius of the circular path followed by the charged particle will be proportional to
1. V/u
 2. u/V
 3. V^2/u^2
 4. u^2/V^2

Options 1. 1
2. 2
3. 3
4. 4

- Q.44 A wire of resistance 4Ω is used to make a coil of radius 7 cm. The wire has a diameter of 1.4 mm and the resistivity of its material is $\rho = 2 \times 10^{-7} \Omega \text{ m}$. The number of turns in the coil will be
1. 70
 2. 40
 3. 140
 4. 20

Options 1. 1
2. 2
3. 3
4. 4

- Q.45 The electric potential due to an electric dipole
- (A) depends on r , where r is the magnitude of position vector $\vec{r}$
- (B) depends on the angle between the position vector $\vec{r}$ and the dipole moment vector $\vec{p}$
- (C) falls off at long distances, as $1/r^2$
- (D) does not depend upon the distance separating the charges
- Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

- Q.46 Which of the following statements are correct?
- (A) A nucleus of mass number A has a radius R given by the expression $R = R_0 A^{1/3}$
- (B) Volume of nucleus is proportional to mass number A
- (C) The density of nucleus increases with the radius of nucleus.
- (D) Density of nuclear matter does not depend on its mass number A
- Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (B), (C) and (D)

3. (A), (B) and (D) only

4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.47 The magnitude of magnetic field inside a solenoid of length $L = 0.3$ m having $N = 800$ turns carrying a current of $I = 6$ A is

1. 2.03 T

2. 60.3 mT

3. 20 mT

4. 6.03 T

Options 1. 1

2. 2

3. 3

4. 4

Q.48 The photoelectric current is directly proportional to the number of photo electrons emitted per second. This implies that

1. the number of photoelectrons emitted per second is equal to the frequency of incident radiation.

2. the number of photoelectrons emitted per second is inversely proportional to the intensity of incident radiation.

3. the number of photoelectrons emitted per second is directly proportional to the intensity of incident radiation.

4. the number of photoelectrons emitted per second is not related to the intensity of incident radiation.

Options 1. 1

2. 2

3. 3

4. 4

Q.49 Two point charges placed a distance d apart in vacuum exert a force of magnitude F on each other. One of the two charges is doubled. To keep the magnitude of force same the separation between the charges should be changed to

1. $2d$

2. $d/2$

3. $\sqrt{2}d$

4. $d/\sqrt{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.50 An electron in the ground state of a hydrogen atom absorbs 12.09 eV energy. The angular momentum of the electron increases by

1. $(h/2\pi)$

2. $2(h/2\pi)$

3. $3(h/2\pi)$

4. $4(h/2\pi)$

Options 1. 1

2. 2

3. 3

4. 4